

WASP-12b

R-1041

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_161221 · created 2026-07-30T08:04:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457742.85443 +/- 0.00034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.052 +/- 0.027
Transit depth (Rp/Rs)^2	0.28 +/- 0.28 [%]
Orbital Inclination (inc)	86.6 +/- 2.85
Airmass coefficient 1 (a1)	1.134 +/- 0.029
Airmass coefficient 2 (a2)	-0.099 +/- 0.016
Scatter in the residuals of the lightcurve fit is	2.55 %
Best Comparison Star	#1 - [83, 75]
Optimal Method	PSF photometry
Transit Duration (day)	0.1201 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.052 ± 0.027 vs 0.1178 (deviation 1.2σ)
Duration fit vs published	0.1201 d vs 0.125 d (deviation 0.5σ)
Mid-time O–C	-1571.0 min
Depth SNR	0.9
Residual scatter	2.55 %

Light curve

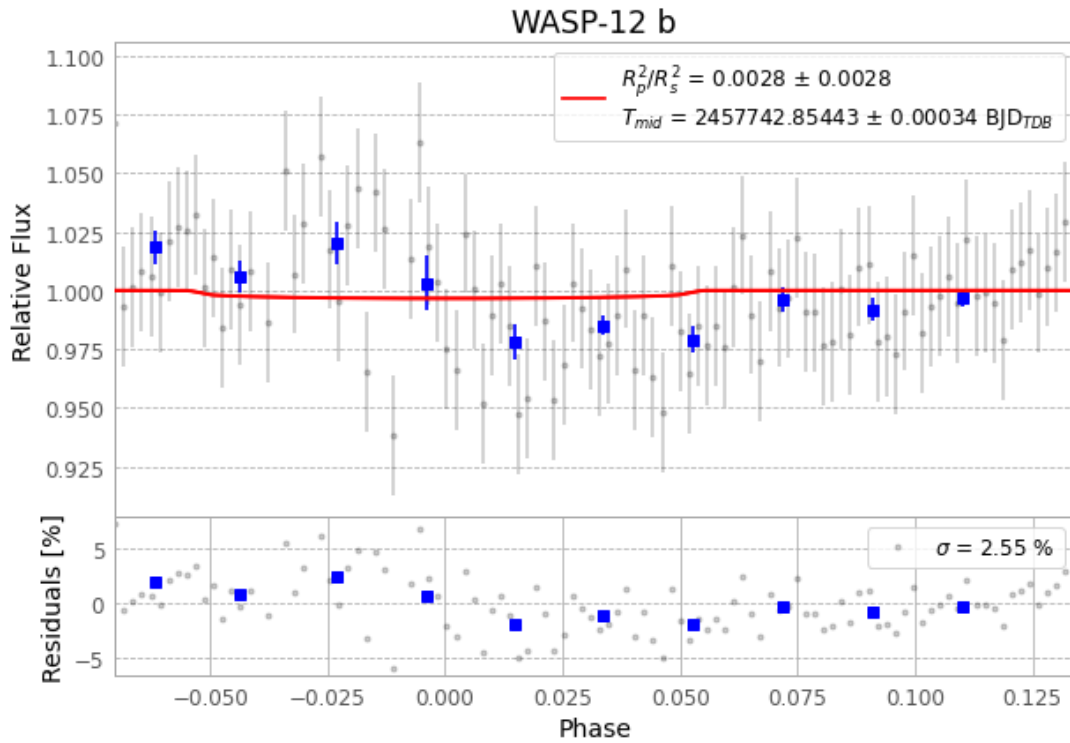


Figure 1 — FinalLightCurve_WASP-12 b_2016-12-19.png (SHA-256 34b6f61f95bc96cf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [418, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cf617353d811ee33...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 77e40fd

Data provenance

108 FITS frames (71 MB), WASP-12161220063012.FITS → WASP-12161220115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-161221.log (552d7025230d...), FinalLightCurve_WASP-12 b_2016-12-19.png (34b6f61f95bc...), FinalLightCurve_WASP-12 b_2016-12-19.csv (575e84335294...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 08:04Z.